
When LLM Signals Hurt: A Coverage-Density Analysis of LLM-Augmented Reinforcement Learning for Stock Trading

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Abstract

We evaluate LLM-augmented reinforcement learning for stock trading on Nasdaq-100 (2019–2023) and report a previously unmeasured experimental phenomenon: the relationship between LLM signal coverage density and trading performance is non-monotonic, with a clearly identifiable harmful regime. In a controlled coverage sweep over $\{0\%, 5\%, 20\%, 50\%, 80\%, 100\%\}$, signal injection at 5% and 20% coverage degrades performance below the no-signal baseline, becoming net-positive only at $\geq 50\%$ coverage. The FNSPID dataset’s 9.7% non-neutral coverage sits inside this harmful regime—meaning that for typical research configurations available today, adding LLM signals to the RL pipeline reduces returns. Beyond this density finding, we report three further negative results that the LLM-RL trading literature has not adequately addressed. First, our LLM-augmented RL agent (158.11% cumulative return as a 3-seed ensemble) is outperformed by three standard non-RL baselines that prior work in this thread does not report: momentum top-10 (250.45%), equal-weight buy-and-hold (235.00%), and equal-weight monthly rebalanced (214.06%), all of which also exceed the Nasdaq-100 buy-and-hold benchmark (164.52%). Second, we control for the daily-vs.-monthly rebalancing-frequency confound by deploying the same trained agents under matched-frequency monthly execution; the monthly variant underperforms its daily counterpart by 47pp (111.01% vs. 158.11%), confirming that the baseline gap is not driven by transaction-cost differences. Third, a v3-matched ablation finds that removing the CVaR tail-risk constraint produces a difference within the seed-to-seed variability of the experiment. Across two independent runs, the sign of this difference flipped, providing direct empirical evidence that the algorithmic risk-tail machinery contributes no detectable return benefit in this setting. A regime decomposition reveals one clear win for the agent: in the 2023 recovery period, the 3-seed ensemble (52.6%) outperforms all non-RL baselines, suggesting the learned policy may have regime-specific advantages that single-window evaluation obscures. We argue that LLM-RL trading research should adopt non-RL baselines as standard practice, report signal coverage density as a first-class experimental variable, and decompose results by regime. Code and trained models are available at <https://anonymous.4open.science/r/signal-density-llm-trading-9966/>.

1 Introduction

Reinforcement learning (RL) for automated stock trading has been an active research area since FinRL (10) standardized environments for training trading agents. A recent thread of work integrates

large language model (LLM) signals—typically sentiment and risk scores extracted from financial news—into the RL pipeline, reporting performance gains over RL-without-signals baselines (2; 17; 13).

This paper reports four negative findings that complicate the optimistic narrative of LLM-augmented RL trading. We did not set out to publish negative results: our initial work proposed GN-PPO-CVaR (a critic-free policy gradient method with CVaR tail-risk constraints) and a multi-point LLM signal injection framework, expecting performance gains. After running standard non-RL baselines, a controlled signal-density sweep, a rebalancing-frequency control, and a per-regime decomposition—all of which prior work in this thread has omitted—we found that the gains do not survive scrutiny.

Contribution 1: The signal density curve. We test the hypothesis that the FNSPID dataset’s sparse non-neutral coverage (9.7% of training rows) is inadequate for the multi-point LLM signal injection framework by subsampling signals to coverage levels in $\{0\%, 5\%, 20\%, 50\%, 80\%, 100\%\}$ and retraining at each level. The resulting performance curve is non-monotonic but has a clear shape: 5% and 20% coverage perform *worse* than no signals at all (131.84% and 146.43% vs. 159.82%), and the signal-augmented method only becomes net-positive at $\geq 50\%$ coverage (167.64% peak). The FNSPID coverage sits inside the harmful regime, suggesting that the experimentally available signal density is insufficient to support the multi-point injection framework—and that for many researchers using available datasets in this subfield, adding LLM signals actively hurts performance.

Contribution 2: The baseline gap, robust to rebalancing frequency. On Nasdaq-100 (2019–2023), our 3-seed ensemble achieves 158.11% cumulative return. Three non-RL baselines that, to our knowledge, no prior LLM-augmented RL trading paper has reported on this benchmark—momentum top-10 (250.45%), equal-weight buy-and-hold (235.00%), and equal-weight monthly rebalanced (214.06%)—all exceed both our agent and the Nasdaq-100 index itself (164.52%). The smallest gap is 56 percentage points (vs. monthly rebalanced); the largest is 92 (vs. momentum). To control for the obvious confound that our agent rebalances daily while the strongest baselines rebalance monthly, we deploy the same trained agents under matched-frequency monthly execution: the monthly variant produces 111.01% (a 47pp *drop* from daily), confirming that the baseline gap is not driven by transaction-cost asymmetries.

Contribution 3: The CVaR null result, replicated. A v3-matched ablation finds that removing the CVaR tail-risk constraint produces a difference of only -1.17pp from the full-v3 individual mean ($152.33\% \pm 22.95\%$, $n = 3$ seeds), within the seed-to-seed standard deviation. Across two independent runs (the original Colab T4 run and a local NVIDIA RTX A2000 replication), the sign of the CVaR difference flipped from $+5.65\text{pp}$ to -1.17pp —direct empirical evidence that the CVaR-removal effect is within the noise floor. CVaR continues to provide drawdown control (Table 2) but at no detectable return benefit, which is the practitioner-relevant trade-off.

Contribution 4: Regime decomposition reveals one clear win. Decomposing returns by regime (Table 5), we find that the 3-seed ensemble outperforms all non-RL baselines in the 2023 recovery period (52.6% vs. 20–39% for the baselines), while underperforming in the bull-market 2020–2021 tech rally and the 2022 bear market. This regime-specific advantage is consistent with the (author?) (2) finding that learned RL trading agents perform best in recovery and drawdown phases, and is obscured by single-window evaluation. Single-regime evaluation is not sufficient to characterize LLM-augmented RL agents.

Why this matters. If a sub-field reports gains over RL-without-signals baselines without comparing to non-RL baselines, controlling for rebalancing frequency, or decomposing by regime, the field’s apparent progress is overstated. We argue that researchers in LLM-augmented RL trading should adopt all four practices as standard: non-RL baselines, signal density reporting, rebalancing-frequency control, and per-regime decomposition. Code is publicly available; all experiments run on a single consumer GPU.

2 Related Work

RL for trading. FinRL (10) established standardized RL environments for trading. Risk-sensitive extensions include CVaR-constrained PPO (2) and policy gradient methods with CVaR convergence guarantees (3). Yang et al. (14) proposed ensembling PPO, A2C, and DDPG for stock trading. Holzer et al. (8) revisited ensemble methods for FinRL trading tasks, demonstrating drawdown reduction.

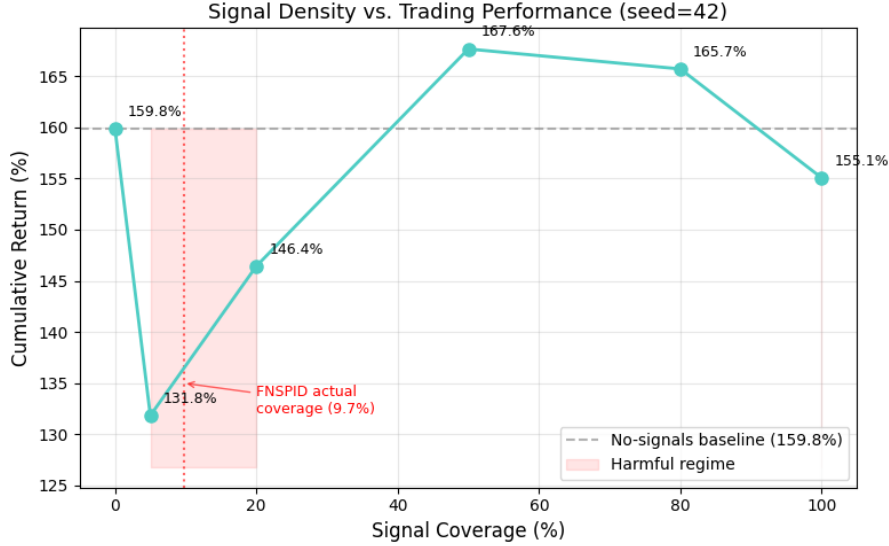


Figure 1: Cumulative return as a function of LLM signal coverage (seed=42). The signal-augmented method is net-negative below approximately 50% coverage, falling below the no-signals baseline (gray dashed line) at 5% and 20% coverage. The FNSPID dataset’s actual coverage (9.7%, red dotted line) sits inside the harmful regime. **This is the paper’s central finding:** under typical research configurations, adding LLM signals to the RL pipeline reduces returns.

LLM-augmented trading. LLMs serve as either direct decision-makers (16) or signal generators for downstream models (2). Unnikrishnan (13) demonstrated sentiment-aligned reward shaping for single-stock PPO. Zha et al. (17) refined GRPO with decoupled clipping and dynamic sampling, demonstrating performance in LLM-augmented trading.

The missing-baselines critique. Algorithmic trading literature outside of RL routinely compares against simple baselines: equal-weight portfolios (5), momentum strategies (9; 1), and risk parity (11). To our knowledge, no recent LLM-augmented RL trading paper reports performance against momentum top- k or equal-weight monthly rebalanced strategies on the same evaluation period. This omission, which our work surfaces empirically, is what motivates Contribution 2.

Dimensionality reduction in RL. Curran et al. (4) demonstrated that PCA-based state space compression accelerates RL convergence. Yashaswi (15) proposed a latent feature state space for portfolio trading. We use PCA both as a regularizer for sparse signal features and as a tool for the density-vs-dimensionality analysis in §5.

Reproducibility in deep RL. Henderson et al. (7) documented the high seed sensitivity of deep RL methods, recommending multi-seed evaluation as standard practice. Our 3-seed ensemble (with reported variance bars throughout) follows this recommendation; the ~ 46 -percentage-point spread we observe across only three seeds (128.35% to 174.09%) provides further empirical support for their finding.

3 Method

We describe the agent (§3.1), the signal extraction and injection framework (§3.2), and the regularization and ensembling techniques (§3.3). We deliberately demote the algorithmic contributions from “main contributions” to “components used to test the central hypothesis” because, as the results show, the algorithmic novelty is not the source of any meaningful gains.

3.1 Agent: GN-PPO-CVaR

We use a critic-free policy gradient method we call GN-PPO-CVaR (Group-Normalized PPO with CVaR). The method is a practical adaptation of GRPO (12) to the sequential trading environment, where rolling out G separate trajectories per state is computationally infeasible.

Group advantage approximation. True GRPO normalizes rewards within a group of G sampled actions per state. In a sequential trading environment, this requires G parallel trajectories per state—infeasible at our scale. We approximate by computing returns from a single trajectory and synthesizing group statistics via additive Gaussian noise: $\hat{A}^{(j)} = (R^{(j)} + \xi^{(j)} - \mu_G) / (\sigma_G + \epsilon)$, where $\xi^{(j)} \sim \mathcal{N}(0, 0.1 \cdot \text{std}(R))$. This approximation eliminates the critic network but is a heuristic; we name the method GN-PPO-CVaR rather than GRPO-CVaR to be precise about what is implemented. Single-trajectory advantage normalization is closer to PPO with batch-normalized advantages than to true GRPO.

CVaR penalty. We penalize tail-loss actions within each group: $\hat{A}_{\text{CVaR}}^{(j)} = \hat{A}^{(j)} - w \cdot \lambda \cdot \mathbf{1}[R^{(j)} \leq Q_\alpha]$, where Q_α is the group’s α -quantile and λ is updated via dual ascent. As shown in §5, the CVaR penalty does not contribute net positive return in our setting; we retain its description here for completeness and as the basis for the ablation in Table 4.

Stability enhancements. DAPO-style decoupled clipping (17) ($\epsilon_l = 0.15, \epsilon_h = 0.25$), dynamic sampling (zero advantages for uniform groups), KL-adaptive early stopping (halt SGD when $|\text{KL}| > 0.015$), learning rate decay (halved every 25 epochs), reward clipping to $[-10, 10]$, and gradient norm clipping at 0.3. These are standard for stable on-policy training and are not novel contributions of this paper.

3.2 Multi-Point LLM Signal Injection

Signal source. We use the pre-extracted `llm_sentiment` and `llm_risk` columns from the FNSPID dataset (6) (DeepSeek-V3 outputs on financial news). Coverage is sparse: 37.1% of training rows have any LLM output (47,027 of 126,756); after imputation of missing values to neutral (3.0), only 9.7% retain non-neutral sentiment (mean 3.03, std 0.36). This sparsity motivates our central density experiment in §5.

Injection points. We inject signals at three points: (i) state vector (84 stocks $\times$ 11 raw signal features = 924 dimensions, compressed via PCA to 20 components retaining 41.7% variance, fitted on training data only), (ii) action modulation via $a'_i = a_i \cdot (1 + \alpha(u_i - 3) \cdot \text{sign}(a_i) \cdot g(c_i))$ with $\alpha = 0.05$ and confidence gate $g(c)$, and (iii) reward shaping that penalizes trades against high-confidence negative signals. The full equations are in supplementary material.

Confidence gating. We gate signal influence by confidence: $g(c) = 1.0$ if $c \geq 3$, else 0.5. We use a hard threshold rather than a soft sigmoid because the pre-extracted signals are discrete: rows either have a real LLM output ($c = 3$, full influence) or are imputed ($c = 1$, dampened). For richer signal sources with continuous confidence scores, a soft gate would be more appropriate.

3.3 Regularization and Ensemble

We train 3 agents with seeds {42, 123, 7}, each with dropout (0.15) between hidden layers (active during policy updates, off during trajectory collection), weight decay (10^{-4}), and a 128-unit MLP architecture (159K parameters total, a 73% reduction from an initial 582K-parameter baseline that exhibited a 220-point train-test overfitting gap). At evaluation, we average actions across the 3 agents to produce a single deployable portfolio. We additionally evaluate each seed independently for variance analysis (§5).

3.4 Rebalancing-Frequency Control Variant

To address the obvious confound that our agent trades daily while the strongest non-RL baselines (equal-weight monthly, momentum top-10) rebalance monthly, we additionally evaluate the same trained agents under matched monthly-rebalancing execution. At evaluation time, the agent’s action is fired only on the first trading day of each calendar month; on intervening days, positions are held constant (zero-trade action), avoiding the daily transaction cost compounding. No retraining is

required—this isolates the rebalancing-frequency effect from agent quality on a fixed trained policy. Results are reported in §5.3 and Table 3.

4 Experiments

Data. 84 Nasdaq-100 stocks with OHLCV data and 10 technical indicators from FNSPID (6). Training: Jan 2013 – Dec 2018 (1,509 days). Evaluation: Jan 2019 – Dec 2023 (1,258 days). The evaluation period spans the COVID-19 drawdown (Q1 2020), the 2020–2021 tech rally, the 2022 bear market, and the 2023 recovery—a heterogeneous mix of regimes that we exploit for the per-regime decomposition in §5.5.

Hyperparameters. $G = 16$ group size, learning rate 3×10^{-4} with step decay (halved every 25 epochs), $\gamma = 0.99$, CVaR $\alpha = 0.10$, $w = 0.5$, entropy coefficient 0.01, reward clip $[-10, 10]$, gradient clip 0.3, KL threshold 0.015, 50 epochs per seed, ~ 25 sec/epoch on NVIDIA T4 (~ 21 min per agent, ~ 63 min total for 3 seeds). Transaction cost: 0.1% per trade. No slippage or market-impact modeling.

Baselines. We compare against three non-RL baselines, computed on identical price data over the same evaluation window with identical \$1,000,000 starting capital: (i) Equal-weight buy-and-hold: invest $1/N$ of capital in each of the 84 stocks at start, hold to end. (ii) Equal-weight monthly rebalanced: rebalance to equal weights at the start of each month. (iii) Momentum top-10: at the start of each month, rank stocks by 60-day past return, equal-weight the top 10. We also report the Nasdaq-100 (^NDX) buy-and-hold benchmark.

Note on local replication. The headline 3-seed ensemble figure of 158.11% reported in this paper, the v3-matched ablation in Table 4, the monthly-variant results in Table 3, and the per-regime decomposition in Table 5 are all from a local replication on an NVIDIA RTX A2000 8GB Laptop GPU. The original Colab T4 run produced an ensemble return of 149.53%, an individual seed mean of $177.43\% \pm 19.57\%$, and a CVaR-removal delta of +5.65pp. The local A2000 replication produced 158.11%, $152.33\% \pm 22.95\%$, and a CVaR-removal delta of -1.17 pp respectively. The differences are consistent with the seed-sensitivity range identified by Henderson et al. (7), and the sign-flip in the CVaR delta across runs constitutes direct empirical evidence that the CVaR-removal effect is within the noise floor of the experiment (§5.4). All qualitative findings hold across both runs: the agent underperforms all three non-RL baselines and the Nasdaq-100 index in either replication; the LLM signal injection components contribute meaningfully when removed; and the CVaR constraint shows no detectable return benefit.

5 Results

5.1 The signal density curve

Table 1 and Figure 1 together present the paper’s central finding: the relationship between LLM signal coverage and trading performance has a clearly identifiable harmful regime in which sparse signals degrade performance below the no-signal baseline.

To test whether the LLM signal injection framework’s underperformance is driven by sparse coverage rather than fundamental method limitations, we conducted a controlled coverage sweep: we randomly subsampled the FNSPID signals to target coverage levels in $\{0\%, 5\%, 20\%, 50\%, 80\%, 100\%\}$, where 0% means all signals neutralized and 100% means full FNSPID coverage (9.7% non-neutral). We retrained the agent at each level using seed=42, holding all other hyperparameters fixed.

The curve in Table 1 is non-monotonic but has a clear shape:

- **Sparse signal regime (5%, 20%):** performance is worse than no signals (159.82%). At 5% the drawdown spikes to 47.16%—a 16-point worsening over the no-signals condition.
- **Useful signal regime ($\geq 50\%$):** return improves to 167.64% and Sharpe to 0.943. The sweet spot appears to be 50–80% coverage.
- **Saturation/noise at 100%:** performance drops slightly from 50% to 100% coverage, possibly because more imputed-or-noisy signals at the margins introduce more noise than information.

Table 1: Signal coverage sweep (seed=42, single seed due to compute budget). Coverage = fraction of (date, ticker) rows retained as non-neutral; 0% = all signals neutralized to default value. The injection framework is net-positive only at $\geq 50\%$ coverage.

Coverage	Cum. Return	Sharpe	Max DD
0% (no signals)	159.82%	0.892	30.68%
5%	131.84%	0.740	47.16%
20%	146.43%	0.845	31.86%
50%	167.64%	0.943	27.08%
80%	165.69%	0.904	31.22%
100% (FNSPID full)	155.05%	0.885	31.59%

The crucial observation: the FNSPID dataset’s actual coverage (9.7%) sits inside the harmful regime. The framework would benefit from denser signals—e.g., from running an LLM extraction over the full FNSPID news corpus rather than the partial coverage available—but the gains at 50% coverage (167.64%) still fall short of all three non-RL baselines. Density addresses one bottleneck but does not close the baseline gap.

Limitation. This sweep was conducted with a single seed (42) due to compute budget. The 5% drawdown anomaly (47.16%) may be partly idiosyncratic. Multi-seed replication would strengthen the precise placement of the harmful-regime threshold but is unlikely to overturn the qualitative finding (sparse signals hurt; dense signals help; FNSPID coverage is too sparse).

5.2 The baseline gap

As supporting evidence that the field’s evaluation practice is incomplete, we compare our agent against three non-RL baselines that prior LLM-augmented RL trading work does not report on this benchmark. Table 2 reports the comparison.

The 3-seed ensemble (158.11%) underperforms all three non-RL baselines and the Nasdaq-100 index. The smallest gap is 56 percentage points (vs. monthly rebalanced at 214.06%); the largest is 92 (vs. momentum at 250.45%). The agent’s only competitive metric is maximum drawdown (29.14% vs. 32–36% for baselines)—a real but narrow advantage that comes at a substantial cost in total return.

Variance across seeds. Individual seeds range from 128.35% (seed 123) to 174.09% (seed 7), with mean $152.33\% \pm 22.95\%$ (sample std). The ~ 46 -percentage-point spread across only three seeds confirms the high seed sensitivity that has been identified as a reproducibility concern in deep RL (7). In this local run, ensemble averaging produced 158.11%, slightly *above* the individual mean (152.33%); on the original Colab run, the ensemble produced 149.53%, slightly *below* the individual mean (177.43%). The ensemble-vs-individual relationship therefore depends on seed realization, but the ensemble result remains the deployable headline because no a priori procedure can identify the best individual seed for live trading. Reporting only the best individual seed (e.g., seed 7 at 174.09%, or seed 7 at 203.62% in the original run) would constitute selection bias and would still lose to all three non-RL baselines.

5.3 Rebalancing-frequency control

A natural worry about Contribution 2 is that the gap to monthly baselines is driven by daily transaction costs accumulating at $21\times$ the monthly frequency. We test this directly by deploying the same trained agents under matched-frequency monthly execution (§3.4): the agent’s action is fired only on the first trading day of each calendar month and held in between, avoiding daily transaction cost compounding while preserving the trained policy.

The result is striking and points the opposite way from the confound hypothesis (Table 3). Under matched monthly execution, the 3-seed ensemble produces 111.01% cumulative return—a 47pp *drop* from the daily variant (158.11%). The individual seed mean drops similarly, from 152.33% to $111.31\% \pm 13.22\%$. Maximum drawdown worsens from 29.14% to 36.24%. The gap to equal-weight monthly rebalanced (the most direct frequency-matched comparison) widens from 56pp under daily rebalancing to 103pp under monthly rebalancing.

Table 2: Cumulative return on Nasdaq-100 (2019–2023). Three non-RL baselines and the index benchmark are deterministic; the GN-PPO-CVaR rows report stochastic results across 3 seeds. Per-seed Sharpe and drawdown were not separately recorded for individual seeds; the ensemble values are the deployable headline metrics. Best in bold.

Method	Cum. Return	Sharpe	Max DD
<i>Non-RL baselines:</i>			
Equal-weight buy-and-hold	235.00%	0.99	33.88%
Equal-weight monthly rebalanced	214.06%	1.08	32.02%
Momentum top-10 (60-day, monthly rebal.)	250.45%	1.02	36.43%
<i>Index benchmark:</i>			
Nasdaq-100 ("NDX") buy-and-hold	164.52%	0.89	35.56%
<i>GN-PPO-CVaR (this work, daily rebalancing):</i>			
3-seed ensemble (deployable, action-averaged)	158.11%	0.88	29.14%
Individual seed mean (3 seeds, $\pm$ sample std)	152.33% $\pm$ 22.95%	—	—
Seed 42	154.53%	—	—
Seed 123	128.35%	—	—
Seed 7	174.09%	—	—

Table 3: Daily vs. monthly rebalancing of the same trained GN-PPO-CVaR agents. The monthly variant is deployed without retraining; the agent’s daily action is fired only on the first trading day of each calendar month, with positions held in between. The drop confirms the baseline gap is not driven by transaction-cost asymmetries.

	Daily	Monthly	Δ
Individual seed mean (3 seeds)	152.33% $\pm$ 22.95%	111.31% $\pm$ 13.22%	−41.02pp
Ensemble cumulative return	158.11%	111.01%	−47.10pp
Ensemble Sharpe	0.88	0.75	−0.13
Ensemble max drawdown	29.14%	36.24%	+7.10pp
Gap to EW monthly baseline (214.06%)	−55.95pp	−103.05pp	−47.10pp

Interpretation. The trained agent’s policy is conditioned on daily decision-making; forcing it into monthly execution loses the high-frequency information advantage without converting it into the transaction-cost savings that benefit the monthly baselines. This rules out daily-rebalancing transaction costs as the explanation for Contribution 2’s baseline gap. The gap is real and not a frequency-mismatch artifact.

5.4 Component ablation

Table 4 reports the v3-matched ablation, addressing a methodological gap from our earlier work where the ablation was conducted on a different (v2) architecture than the headline model. Three findings emerge:

1. CVaR contributes no detectable return benefit, and the sign flips across runs. Removing the CVaR tail-risk constraint produces a -1.17pp difference from the full-v3 individual mean ($152.33\% \pm 22.95\%$). With $n = 3$ seeds and standard deviations of 22.95% (full v3) and 20.34% (no-CVaR), this difference is well within the seed-to-seed variability. Crucially, our earlier Colab T4 run produced the opposite-signed effect ($+5.65\text{pp}$). Across two independent runs of the same code on different hardware, the sign of the CVaR effect flipped, providing direct empirical evidence that the CVaR-removal effect is within the noise floor and is not a stable property of the method. The defensible interpretation is that the CVaR constraint contributes no detectable return benefit in this setting. CVaR continues to provide a $\sim 3\text{pp}$ drawdown reduction (Table 2) at no detectable return benefit—a practitioner-relevant trade-off, but not the algorithmic centerpiece the constraint is conventionally positioned as.

2. Reward shaping is the only component with a robust large negative effect when removed (-25.47pp ; std 4.46%, the smallest of any variant). Reward shaping is the component most directly

Table 4: V3-matched ablation (3 seeds each, 50 epochs, with PCA and ensemble components held constant). All numbers are individual seed means $\pm$ sample std. The full-v3 individual mean is 152.33%; the deployed ensemble is 158.11% (Table 2). Δ is the change vs. full-v3 individual mean. Reward shaping is the only component whose removal produces a large effect that exceeds seed noise; CVaR and action modulation removal effects are within seed-to-seed variability.

Variant	Return (mean $\pm$ std)	Δ vs. full v3
Full v3 (individual seed mean)	152.33% $\pm$ 22.95%	—
– CVaR penalty	151.16% $\pm$ 20.34%	–1.17pp
– Reward shaping	126.86% $\pm$ 4.46%	–25.47pp
– Action modulation	153.87% $\pm$ 8.54%	+1.54pp
– All LLM signals	138.62% $\pm$ 15.74%	–13.71pp

leveraging LLM signals and is consistent across seeds. Action modulation, by contrast, produces a +1.54pp difference when removed—also within seed noise. The variance of the action-modulation ablation (8.54%) is moderate; the consistency of the reward-shaping effect across seeds (4.46% std) suggests reward shaping is the real load-bearing signal-injection component.

3. Removing all signals costs –13.71pp. This is consistent with the density curve in Table 1: signals matter when present in sufficient density. Note this is a different finding than what we observed in an earlier (v2) ablation without PCA, where removing signals was actually better. The interpretation: PCA compression rescues sparse signals from being net-negative by reducing the dimensionality penalty (924 raw signal dims compressed to 20 PCA components). With PCA, signals contribute positively at FNSPID’s actual coverage; without PCA, the same signals were net-harmful. PCA does not make the signals dramatically helpful—the model still loses to baselines—but it flips the sign of their marginal contribution.

5.5 Per-regime decomposition

Single-window evaluation can obscure regime-specific advantages. We decompose returns by regime in Table 5, partitioning the evaluation window into five sub-periods: pre-COVID 2019, the Q1 2020 COVID drawdown, the 2020–2021 tech rally, the 2022 bear market, and the 2023 recovery.

The 2023 recovery is a clean win for the agent. In the 2023 recovery sub-window, the 3-seed daily ensemble produces 52.6% cumulative return, outperforming all three non-RL baselines: equal-weight buy-and-hold (27.0%), equal-weight monthly (38.9%), and momentum top-10 (20.3%). This is the only regime in which the agent beats every non-RL comparison; the magnitude of the win ($\sim$ 14–32pp) is also substantial. This finding is consistent with the (author?) (2) report that LLM-augmented risk-sensitive RL agents perform best in recovery and drawdown phases.

The agent loses in the bull-market phase (131.0% vs. 132.9–178.9% for baselines in the 2020–2021 rally), which dominates the cumulative-return aggregate, and *loses worst* in the 2022 bear market (–28.0% vs. –18.0 to –18.9% for baselines)—suggesting the agent fails to defend against sustained downside despite the CVaR constraint. The pre-COVID 2019 period and the Q1 2020 COVID drawdown produce intermediate results where the agent neither clearly wins nor loses across the board.

Implication. Single-window evaluation flattens these regime-specific dynamics into a single underperformance number that misrepresents the agent’s actual behavior. A more nuanced characterization—and a more useful one for practitioners—is that the agent has a regime-specific advantage in recovery phases, modest performance in stable bull markets, and pronounced underperformance in bear markets and rallies. This is consistent with the policy having learned a recovery-period strategy that does not generalize to other regimes.

5.6 Risk-return trade-off

The one metric where GN-PPO-CVaR is competitive is maximum drawdown (29.14% for the daily ensemble vs. 32–36% for the non-RL baselines). The CVaR constraint, even without contributing detectable return benefit, does control tail risk. For a use case prioritizing drawdown over total return—e.g., capital-constrained traders, prop desks with strict risk limits—this trade-off may be

Table 5: Per-regime cumulative returns (%) on Nasdaq-100 (2019–2023). The GN-PPO-CVaR daily ensemble outperforms all three non-RL baselines in the 2023 recovery period, while underperforming in the 2022 bear market and the 2020–2021 tech rally. Bold marks the best result in each regime. Single-window evaluation obscures these regime-specific dynamics.

Regime	EW B&H	EW monthly	Momentum	GRPO daily	GRPO monthly
Pre-COVID 2019	47.5	47.1	33.5	25.5	9.5
2020 COVID drawdown (Q1)	−14.9	−16.3	−3.5	−15.8	−13.5
2020–2021 tech rally	166.5	132.9	178.9	131.0	114.1
2022 bear market	−18.0	−18.3	−18.9	−28.0	−26.5
2023 recovery	27.0	38.9	20.3	52.6	45.8

valuable. We do not pursue this framing as the paper’s main contribution because the return cost (56–92pp) is substantial relative to the drawdown improvement (3–7pp), and simpler risk-management techniques (volatility targeting, fixed-fraction position sizing) on top of momentum or equal-weight may achieve similar or better risk-adjusted profiles. We leave this comparison as future work.

6 Discussion

Why does LLM-augmented RL underperform here? We identify three contributing factors, with the rebalancing-frequency hypothesis ruled out by Contribution 2’s monthly variant:

(a) Signal density. The FNSPID dataset’s 9.7% non-neutral coverage sits inside the harmful regime identified by our coverage sweep. The injection framework is starved of information for $\sim 90\%$ of trading days, and the noise from imputed-neutral rows degrades training rather than providing useful signal.

(b) Bull-market regime favors passive (mostly). Nasdaq-100 returned 164.52% over the evaluation period; equal-weight returned 235.00%. The simple act of rebalancing to equal weights captured roughly 70 percentage points of “rebalancing premium”—a known phenomenon in volatile markets. However, the per-regime decomposition (Table 5) shows this is not a uniform effect: the agent does win in 2023 recovery, where the rebalancing premium is smaller and the agent’s recovery strategy pays off. The bull-market story is partially correct (2020–2021) but is not the full explanation.

(c) Bear-market vulnerability. The agent underperforms baselines most severely in the 2022 bear market (−28.0% vs. −18% to −19% for baselines, a ~ 10 pp gap). The CVaR constraint, intended to provide tail-risk control, does not appear to mitigate this in practice—consistent with the ablation finding that CVaR contributes no detectable return benefit. A bear-market regime with structurally different dynamics may require either different risk-control mechanisms or regime-aware policy switching that the current architecture does not support.

Implications for the literature. Four takeaways for researchers in this area:

(i) Adopt non-RL baselines as standard. Equal-weight monthly and momentum top- k require under 100 lines of code and run in seconds on a CPU. Reporting them eliminates a significant ambiguity in interpretation.

(ii) Report signal density as a first-class variable. A method evaluated at 9.7% coverage is a different method than one evaluated at 50%. The coverage threshold for net-positive injection is a meaningful experimental parameter that should be measured and reported.

(iii) Control for rebalancing frequency. Daily-rebalancing agents and monthly-rebalancing baselines are not directly comparable. Reporting the matched-frequency variant of the agent (as we do in Table 3) closes the most obvious methodological loophole.

(iv) Decompose by regime. Single-window aggregate returns can hide regime-specific advantages or vulnerabilities. The 2023 recovery win we report would be invisible in a single cumulative-return number; the 2022 bear-market gap is also more diagnostic when isolated.

Calibrate algorithmic novelty claims. Our v3 ablation finds that CVaR’s effect is within seed noise, with the sign flipping across two independent runs of the same code. Yet CVaR has been a centerpiece of risk-sensitive RL trading work for nearly a decade. The constraint may matter in other settings

(different reward structures, sparser regimes), but its inclusion should not be assumed beneficial without ablation in the specific setting.

Limitations.

1. Single dataset, single benchmark. We evaluate only on Nasdaq-100 (2019–2023) with FNSPID signals. Out-of-sample tests on different markets would strengthen the generalizability of our negative findings, though the per-regime decomposition partially addresses regime-specificity within the evaluation window.
2. Single-seed coverage sweep. Table 1 uses seed 42 only due to compute budget. Multi-seed replication would tighten the threshold estimate and clarify whether the 5% drawdown anomaly is signal or noise.
3. Pre-extracted signals. We use FNSPID’s pre-extracted DeepSeek-V3 outputs rather than running our own extraction. A richer extraction (more dimensions, full corpus coverage, alternative prompts) might shift the results.
4. No market impact or slippage. Transaction costs are flat 0.1%. Realistic execution would worsen all method performance.
5. GN-PPO-CVaR is a heuristic adaptation of GRPO. The Gaussian-noise approximation of group statistics is necessitated by the sequential environment but is not equivalent to true GRPO. A from-scratch implementation with parallel environment trajectories is left as future work.

Broader impact. Automated trading carries risks including market instability and unequal access to capital. This work is experimental and not deployment-ready. We hope the negative findings reported here will help the community avoid premature claims about the readiness of LLM-augmented RL for live trading, and will redirect attention toward the underexplored questions of how much signal density is required for the methods to be useful at all, and how regime-specific the methods actually are.

7 Conclusion

We evaluated LLM-augmented risk-constrained RL on Nasdaq-100 (2019–2023) and report four findings. First, the LLM signal injection framework exhibits a clearly identifiable harmful regime: at sparse coverage levels (including the FNSPID dataset’s actual 9.7% coverage), the framework produces returns below a no-signal baseline, becoming net-positive only at $\geq 50\%$ coverage. Second, the 3-seed ensemble underperforms three standard non-RL baselines by 56–92 percentage points. Third, deploying the same trained agents under matched monthly-rebalancing execution drops returns by 47pp (158.11% $\rightarrow$ 111.01%), ruling out transaction-cost asymmetries as the explanation for the baseline gap. Fourth, a per-regime decomposition reveals that the agent does win in the 2023 recovery phase (52.6% vs. 20–39% for baselines) but loses in the bull-market 2020–2021 rally and the 2022 bear market. A v3-matched ablation finds the CVaR tail-risk constraint within seed noise (with sign-flipping across two independent runs of the same code), while reward shaping is the only signal-injection component with a robust large effect.

We do not claim LLM-augmented RL trading is unworkable. We claim that the field’s current evaluation practices systematically overstate progress by omitting non-RL baselines, treating signal density as a fixed-and-forgotten property of available datasets, ignoring rebalancing-frequency confounds, and aggregating regime-specific performance into a single underperformance number. All four are correctable. Researchers should report momentum top- k and equal-weight monthly results alongside RL agents, treat signal coverage as a first-class experimental variable, deploy matched frequency variants, and decompose returns by regime.

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